

The γ chain

b is μ -Parseval of the border; w_k is not uniform; $\gamma < 1$ stays a census

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August 29, 2026

Abstract

Round 423 left $\gamma < 1$ as the missing chain step for $\text{den} < 2$. The border vector b is the coefficient vector of the signed border measure σ in the μ -orthonormal frame: $b_k = \langle \sigma, \hat{\pi}_k^\mu \rangle$, and $\|b\|^2$ equals the μ -side Parseval budget. $\|b\|^2 \leq B_w$ compares this energy to a *different* chain (the μ -tilde ρ -budget plus $\frac{5}{7}$). Termwise $w_k = b_k^2/\rho_k$ is not bounded by a uniform $\bar{w} < 1$. On the named census $\|b\|^2 \leq 0.80 S$ (so the $\frac{5}{7}$ buffer is not needed) and $\gamma \leq 0.724$. No RH claim.

Status. Lemma ?? is **reduced**. Proposition ?? is a finite theorem. Ausgang BESSEL_EXACT / W_TERM_REFUTED / SHARP_CENSUS / GAMMA_OPEN. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The lemma

Lemma 1 (γ chain). *On the border chain, $\|b\|^2 \leq B_w = S_{N-2} + \frac{5}{7}$ (or $\|b\|^2 \leq q B_w$ with an explicit $q < 1$), so $\gamma < 1$ is a theorem and $\text{den} = 1 + \gamma - v_t \cdot s < 2$ follows even at $v_t \cdot s = 0$.*

Verdict: REDUZIERT. b is identified (Proposition ??). A uniform termwise bound $w_k \leq \bar{w} < 1$ is refuted. The sharpest holding form on the census is $\|b\|^2 \leq 0.80 S$, which is stronger than $\|b\|^2 \leq B_w$ but is not proved. $\gamma < 1$ as a chain theorem remains open.

2 What b is

Proposition 1 (μ -Parseval of the border). *Let $\{\hat{\pi}_k^\mu\}$ be the μ -orthonormal OP frame (three-term recurrence of the positive zone, start $1/\sqrt{h_0^\mu}$). The Sherman–Morrison border column is*

$$b_k = \sum_a w_a \hat{\pi}_k^\mu(t_a) = \langle \sigma, \hat{\pi}_k^\mu \rangle,$$

where σ is the signed border measure (border atoms of μ with weights w , border atoms of ν with weights $-v$). Parseval on the truncated frame is exact: $\|b\|^2 = \sum_{k \leq N} (\int \hat{\pi}_k^\mu d\sigma)^2$, equal to the μ -side budget. Over $\mathbb{Q}$, $b = (\frac{3}{5}, \frac{4}{5})$ with leftover $\frac{7}{9}$ gives $\|b\|^2 = 1 < \frac{16}{9} = \|\varphi\|^2$. At $k_z = 9$, $\|b\|^2 = 5.6813$ matches the μ -side budget to $4 \cdot 10^{-14}$.

Thus a same-frame Bessel $\|b\|^2 \leq \|\varphi_B\|_{\text{chain}}^2$ is true with $\varphi_B = \sigma$ and with the μ -norm on the left. The budget $B_w = S_{N-2} + \frac{5}{7}$ is the partial energy of the μ -tilde chain ($\rho_k = F_k^2/h_k$). $\gamma < 1$ is therefore a cross-chain comparison, not an identity of one inner product.

3 The ratio $w_k = b_k^2/\rho_k$

If $w_k \leq \bar{w} < 1$ uniformly then $\|b\|^2 \leq \bar{w} S < B_w$ at once. This fails: on every core window about 40% of the terms have $w_k > 1$, and the maxima are 10^6 – 10^9 (they sit on near-null ρ steps). At $k_z = 9$, $w_0 = 0.939$, median 0.623, $\text{frac}(w > 1) = 0.43$, $w_{\max} = 1.6 \cdot 10^5$. The sum still closes because large w_k carry tiny ρ_k . A weighted Cauchy–Schwarz of the two chains is the remaining termwise language; a uniform $\bar{w}$ is not.

4 The $\frac{5}{7}$ buffer and the sharp census

Dropping $\frac{5}{7}$ still leaves $\|b\|^2 < S$ on every measured world (selected max $\|b\|^2/S = 0.799$ at $k = 5$; core-42 max 0.766). The buffer is not what saves the inequality on the census. The sharpest holding form is

$$\|b\|^2 \leq 0.80 S < S < S_{N-2} + \frac{5}{7} = B_w.$$

Core-42: $\gamma \in [0.548, 0.724]$. EXT 6/6 the same pattern. $k = 8$ is pinned ($\gamma = 0.6386$), not re-gated.

5 Kills

world / mutant	$\gamma < 1$?	note
core-42 + EXT + selected	yes	census, max 0.724
dead χ (C3-23, C4-20)	yes	also $\ b\ ^2 < S$
atom reorder (rev x)	yes, equal	measure, not order
scramble / shuffle w	n/a	tilde chain fails
unnorm b (drop $1/\sqrt{h_0}$)	no (2.51)	orthonormality is load-bearing
drop $\frac{5}{7}$	yes	max $\ b\ ^2/S = 0.80$

Dead χ keep $\gamma < 1$. Wherever the two chains exist, $\gamma < 1$ does not use the von Mangoldt arithmetic — a candidate chain theorem without arithmetic, still unproved. The scramble world does not supply a counter-example; it never forms B_w .

6 What is a theorem, what is a census

Proposition ?? is a theorem. Lemma ?? is not: $\gamma < 1$ is a cross-chain census, w_k has no uniform bound, and $q = 0.80$ is measured, not derived. If $\gamma < 1$ is proved, $\text{den} < 2$ is immediate ($1 + \gamma < 2$). No RH claim.